

Image-Based Nutrient Deficiency Detection for Onion Crops Using Lightweight CNN Models

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Project Proposal Report

IT22142528

Supervised By: Dr. Kaushalya Dissanayake

Co-Supervised By: Dr. Dharshana Kasthurirathna

B.Sc. (Hons) in Information Technology
Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another

person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
	IT22142528	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

ABSTRACT

Onion (*Allium cepa*) cultivation is a cornerstone of Sri Lankan agriculture, yet farmers face persistent challenges in accurately diagnosing nutrient deficiencies such as nitrogen (N), phosphorus (P), and potassium (K) stress. Visual symptoms of nutrient deficiency—including pale green leaves, tip yellowing, stunted growth, and weak bulbs—frequently overlap with signs of fungal disease and pest damage, making field-level diagnosis unreliable and often leading to excessive or incorrect fertilizer application. Current diagnostic methods depend on costly and time-consuming soil laboratory testing, which is inaccessible to the majority of smallholder farmers.

This research proposes the design, development, and validation of an image-based nutrient deficiency detection system as a core component within an Integrated AI Platform for Onion Crop Health Monitoring. The system employs lightweight CNN architectures—MobileNetV3 and EfficientNet-B0—fine-tuned on a curated dataset of nutrient-deficient and healthy onion leaf images collected from Field Crops Research and Development Institute (FCRDI) partner farms. Feature engineering incorporates colour indices (RGB/HSV), texture measures, canopy density, and growth-stage indicators. Hyperparameter optimization is performed using Bayesian tuning with cross-validation, and model interpretability is ensured through Grad-CAM visualisations and SHAP explainability reports. The system is optimised for mobile deployment

using TensorFlow Lite, enabling offline, lab-free nutrient analysis directly from smartphone-captured leaf images.

The expected outcome is a mobile-deployable model that identifies N, P, K, Mg, and Ca deficiencies with quantitative stress scores, produces actionable fertilizer recommendations tailored to smallholder economics, and reduces over-fertilization by an estimated 25–30%. The system contributes to sustainable onion farming by replacing guess-based fertilizer use with data-driven nutrient management.

Keywords: Nutrient Deficiency Detection, CNN, MobileNetV3, EfficientNet, Onion Crop, Image Classification, Precision Agriculture, TinyML

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
CNN	Convolutional Neural Network
N	Nitrogen

P	Phosphorus
K	Potassium
Mg	Magnesium
Ca	Calcium
RGB	Red, Green, Blue (colour model)
HSV	Hue, Saturation, Value (colour model)
GDD	Growing Degree Days
SHAP	SHapley Additive exPlanations
Grad-CAM	Gradient-weighted Class Activation Mapping
TFLite	TensorFlow Lite
AutoML	Automated Machine Learning
TinyML	Tiny Machine Learning
FCRDI	Field Crops Research and Development Institute
DOA	Department of Agriculture
FAO	Food and Agriculture Organization
SDG	Sustainable Development Goal
IPM	Integrated Pest Management
LSTM	Long Short-Term Memory Network
API	Application Programming Interface
UI	User Interface
WBS	Work Breakdown Structure

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1. INTRODUCTION

1.1 Background and Context

Onion (*Allium cepa*) cultivation is one of the most economically significant agricultural activities in Sri Lanka, contributing substantially to both national food security and the livelihoods of rural farming communities. According to the Department of Agriculture Sri Lanka, onion is cultivated across multiple agro-ecological zones, with the Maha Illuppallama region being a primary production hub. Despite its importance, onion farmers face persistent and multifaceted challenges that severely impact crop health and economic returns.

The Real-World Problem:

Nutrient deficiencies in onion crops—including nitrogen (N), phosphorus (P), potassium (K), magnesium (Mg), and calcium (Ca)—are among the most common yet difficult-to-diagnose field-level problems. Nitrogen deficiency manifests as pale green leaves and stunted growth; potassium deficiency causes yellowing leaf tips and weak bulbs; phosphorus deficiency leads to dark purplish discoloration and delayed maturity [1]. These visual symptoms frequently overlap with signs of fungal diseases (e.g., purple blotch caused by *Alternaria porri*) and pest damage (e.g., Thrips feeding), making accurate field diagnosis extremely challenging through subjective visual inspection alone [2]. Misdiagnosis leads to inappropriate fertilizer application—either excessive (causing soil degradation and water pollution) or insufficient (reducing crop yield and quality).

Who Is Affected and Why It Matters:

Sri Lanka has an estimated 150,000+ smallholder onion farming households. These farmers rely primarily on experience-based "guess work" for fertilizer decisions or on costly and time-consuming soil laboratory testing that is inaccessible to the majority of smallholder farmers. The Food and Agriculture Organization (FAO) reports that improper nutrient management can reduce onion yields by 20–40% and that excessive fertilizer use is a leading contributor to agricultural environmental damage in developing countries [3].

Statistical Evidence:

- Sri Lanka imports approximately 200,000 metric tons of onions annually, highlighting a significant domestic production shortfall attributable in part to suboptimal nutrient management [4].
- Studies indicate that proper nutrient management could increase onion yields by 25–35% in the South Asian context [5].
- Over-fertilization contributes to an estimated 15–20% increase in production costs for smallholder onion farmers [3].
- Only 12% of Sri Lankan farming households have access to digitalized agricultural advisory services [6].

Analysis of Existing Solutions and Their Limitations:

Current approaches to nutrient deficiency diagnosis include manual soil testing, laboratory-based analysis, and generic commercial plant health scan applications. Manual soil testing requires field samples to be sent to laboratories—a process that is expensive (LKR 2,000–5,000 per test), time-consuming (7–14 days for results), and inaccessible for most smallholder farmers. Commercial apps like Plantix and Plant.id offer generic health scan features but do not convert onion-specific symptoms to quantitative N/P/K metrics, lack support for Sri Lankan onion varieties, and provide only English-language outputs without localised fertilizer recommendations suited to smallholder economics [7]. Research on image-based nutrient detection using CNNs and semantic segmentation has shown promise, but onion-specific models under Sri Lankan field conditions are virtually nonexistent [5].

The Proposed Solution:

This research proposes an image-based nutrient deficiency detection system as a core component within an "Automated Diagnostic

Framework for Onion Crop Health Monitoring." The system employs lightweight CNN models (MobileNetV3, EfficientNet-B0) trained on onion leaf images collected from FCRDI partner farms, augmented with colour indices, texture measures, and growth-stage indicators. The model delivers quantitative nutrient stress scores and lab-free fertilizer prescriptions optimised for smallholder economics. The overall platform comprises four integrated components:

1. **Component 1:** CNN-Based Purple Blotch Disease Detection (Member 1 – IT22054890)
2. **Component 2:** Pest Detection & Identification using YOLOv8 (Member 2 – IT22226464)
3. **Component 3:** Multilingual Agro-Advisory Chatbot – Sinhala/English/Singlish (Member 3 – IT22087256)
4. **Component 4:** Image-Based Nutrient Deficiency Detection (Member 4 – IT22142528) (*This component*)

SDG Alignment:

This project directly contributes to:

- **SDG 2 (Zero Hunger):** By optimizing nutrient management to improve crop yield and quality.
- **SDG 12 (Responsible Consumption and Production):** By reducing excessive and inappropriate fertilizer use through data-driven recommendations.
- **SDG 13 (Climate Action):** By minimizing soil degradation and water pollution caused by over-fertilization.

Research Cluster Alignment:

This research is aligned with the Autonomous Intelligent Machines and Systems (AIMS) research cluster and the Agri-Technology industry vertical, focusing on intelligent perception, decision-making, and human-centric AI solutions.

1.2 Literature Review

A comprehensive literature review was conducted examining recent scholarly works in image-based plant nutrient analysis, CNN

architectures for agricultural image classification, feature engineering for leaf analysis, and mobile-deployable machine learning models. The following key findings emerged:

1. CNN-Based Nutrient Deficiency Detection in Crops:

Montaze and Rahimi (2018) demonstrated the feasibility of image-based detection of nutrient deficiencies in onion crops using machine learning, extracting RGB and HSV colour features to classify nitrogen and potassium stress, achieving accuracies exceeding 88% under controlled laboratory conditions [5]. However, their study did not consider the overlapping visual symptoms between nutrient deficiencies and disease/pest damage under field conditions.

2. Deep Learning for Leaf-Based Diagnosis:

Mohanty et al. (2016) applied deep CNNs to the PlantVillage dataset for disease detection, demonstrating that transfer learning from ImageNet-pretrained models significantly improves classification accuracy on small agricultural datasets [8]. Their work established the baseline approach for transfer learning in agricultural image classification, though it was limited to disease detection rather than nutrient analysis.

3. Colour and Texture Feature Engineering:

Guo et al. (2020) developed a multi-feature plant stress detection system combining colour histogram features, texture descriptors (GLCM), and canopy density metrics, achieving superior classification performance compared to single-feature approaches [9]. Their approach demonstrated that engineered features combined with CNN-extracted features provide more robust classification under variable field conditions.

4. Lightweight CNN Architectures for Mobile Deployment:

Howard et al. (2019) introduced MobileNetV3, designed specifically for mobile and edge devices, achieving state-of-the-art accuracy with significantly reduced computational cost [10]. Tan and Le (2019) proposed EfficientNet, demonstrating that systematic scaling of network width, depth, and resolution achieves better accuracy-efficiency trade-offs [11]. Both architectures are well-suited for real-time agricultural applications on smartphones.

5. Semantic Segmentation for Nutrient Analysis:

Chen et al. (2021) applied semantic segmentation to quantify chlorophyll content and nutrient status from crop canopy images, linking visual indices to yield predictors via LSTM networks [12]. Their work demonstrated the potential for pixel-level nutrient analysis but was computationally expensive and unsuitable for mobile deployment without significant optimization.

6. Explainable AI in Agricultural Applications:

Selvaraju et al. (2017) introduced Grad-CAM (Gradient-weighted Class Activation Mapping), enabling visual explanations of CNN predictions by highlighting image regions that most influence classification decisions [13]. Lundberg and Lee (2017) proposed SHAP (SHapley Additive exPlanations) for model-agnostic feature importance analysis [14]. Both techniques are critical for building farmer trust in AI-driven nutrient recommendations.

7. Existing Products and Their Limitations:

Commercial applications like Plantix, Plant.id, and Cropwise offer AI-driven crop health scanning but are fundamentally limited in nutrient deficiency detection. They provide generic health assessments without quantitative N/P/K metrics, lack Sri Lankan onion variety-specific models, operate only in English, and do not provide fertilizer application guidance calibrated to smallholder field sizes and budgets [7].

8. AutoML and Hyperparameter Optimization:

Wang et al. (2021) demonstrated that AutoML-driven feature selection and Bayesian hyperparameter optimization significantly improve model performance on agricultural classification tasks compared to manual tuning, particularly for distinguishing overlapping symptoms [15]. This approach is critical for handling the symptom overlap challenge in nutrient deficiency detection.

1.3 Research Gap

The comprehensive literature review reveals several critical gaps that this research aims to address:

1. Absence of Onion-Specific Nutrient Deficiency Models for Sri

Lankan Conditions:

While CNN-based nutrient detection has been explored for crops like rice, maize, and tomato, no validated model exists that is specifically trained on Sri Lankan onion varieties under tropical field conditions. Existing models trained on generic or foreign datasets fail to account for local climate variations, soil types, and cultivar-specific symptom presentations [5].

2. Inability to Distinguish Nutrient Deficiency from Disease/Pest Damage:

Visual symptoms of nutrient deficiency frequently overlap with disease (e.g., purple blotch) and pest damage (e.g., Thrips feeding marks). Existing commercial apps and research models treat these as independent classification problems rather than addressing the multi-class, overlapping symptom challenge [7][9].

3. No Quantitative Nutrient Stress Scoring System:

Current approaches provide only categorical classification (e.g., "nitrogen deficient" or "healthy") without quantitative severity metrics. No existing system delivers N/P/K/Mg/Ca stress scores that can be directly mapped to fertilizer application guidance [5].

4. Absence of Lab-Free, Smartphone-Accessible Nutrient Analysis:

Soil laboratory testing remains the gold standard for nutrient analysis but is costly, slow, and inaccessible for smallholder farmers. No existing mobile application provides image-based nutrient analysis that can replace or complement soil testing for onion crops [3][7].

5. Lack of Fertilizer Recommendations Optimized for Smallholder Economics:

Existing systems, when they provide fertilizer recommendations, base them on optimal agronomic rates without considering the economic constraints of smallholder farmers—field size, budget limitations, and locally available fertilizer products [3].

Evidence-Based Justification for Novelty:

Feature	Plantix	Plant.id	Cropwise	Lab Soil Test	Proposed System

Onion-Specific N/ P/K Detection	x	x	x	✓ (lab only)	✓
Sri Lankan Variety Validation	x	x	x	✓	✓
Quantitative Stress Score	x	x	x	✓	✓
Overlapping Symptom Differentiation	x	x	x	N/A	✓
Mobile/Offline Inference	x	x	x	x	✓
Smallholder- Calibrated Recommendations	x	x	x	x	✓
Lab-Free Analysis	✓ (generic)	✓ (generic)	✓ (generic)	x	✓ (onion- specific)

Table 1: Comparison of Existing Nutrient Deficiency Detection Systems

2. OBJECTIVE

2.1 Main Objective

To design, develop, and validate an image-based nutrient deficiency detection system that uses lightweight CNN models (MobileNetV3 / EfficientNet-B0) trained on Sri Lankan onion leaf images to identify and quantify N, P, K, Mg, and Ca deficiencies, delivering quantitative stress scores and actionable fertilizer recommendations through a mobile-deployable model, with a target classification accuracy of $\geq 88\%$ validated using multi-season data from 40–50 FCRDI partner farms.

2.2 Specific Objectives

1. **Compile a Multi-Season Nutrient Deficiency Dataset:** Collect and curate leaf images and field data from 40–50 FCRDI partner onion farms covering seasons 2024–2026, including variables such as variety, planting date, daily weather, irrigation and fertilizer logs, disease/pest indices, and image-based nutrient stress indicators.
2. **Engineer Discriminative Visual and Contextual Features:** Develop a comprehensive feature set combining colour indices (RGB/HSV histograms), texture measures (GLCM descriptors), canopy density and growth-stage indicators, and basic management information (variety, fertilizer history) to enable robust classification under variable field conditions.
3. **Train and Optimise Lightweight CNN Classification Models:** Implement and compare MobileNetV3 and EfficientNet-B0 architectures, fine-tuned on the curated dataset using transfer learning from ImageNet, with Bayesian hyperparameter optimization and cross-validation achieving a target F1 score of $\geq 85\%$.
4. **Develop Quantitative Nutrient Stress Scoring:** Create a regression/classification pipeline that maps leaf image features to quantitative N/P/K/Mg/Ca stress scores (0–100%) rather than simple categorical labels, enabling precise fertilizer recommendation mapping.
5. **Implement Model Interpretability Visualisations:** Integrate Grad-CAM and SHAP explainability techniques to highlight which leaf regions and features drive nutrient deficiency predictions, building farmer and expert trust in the AI system.
6. **Optimise for Mobile Deployment:** Compress and convert the trained model to TensorFlow Lite format (<10 MB) through INT8 quantization and pruning, enabling offline inference on basic Android smartphones.
7. **Validate with Stakeholders in the Field:** Conduct field validation

with FCRDI agronomists and 20+ onion farmers across at least three cropping cycles, measuring classification accuracy, prediction reliability, and user acceptance.

3. METHODOLOGY

Research Approach

This research follows a **Design Science Research (DSR)** methodology, which emphasizes the iterative design, development, and evaluation of an IT artifact (the nutrient deficiency detection model) to solve an identified real-world problem. The DSR framework guides the process through problem identification, solution design, development, demonstration, and evaluation cycles [16].

Technical Methods and Algorithms

1. Image Preprocessing and Feature Extraction:

- **Leaf Segmentation:** Automated background removal using GrabCut algorithm and Otsu's thresholding to isolate onion leaf regions from complex field backgrounds
- **Colour Feature Extraction:** RGB and HSV histogram analysis, mean channel values, and vegetation indices (Excess Green Index, Normalized Difference Index)
- **Texture Feature Extraction:** Gray-Level Co-occurrence Matrix (GLCM) descriptors—contrast, correlation, energy, and homogeneity
- **Canopy Analysis:** Canopy density estimation and leaf area measurement from overhead images
- **Data Augmentation:** Rotation (0–360°), random zoom (0.8–1.2×), horizontal/vertical flip, brightness variation ($\pm 20\%$), and Gaussian noise injection

2. CNN Architecture and Training:

A pre-trained lightweight CNN will be fine-tuned on the curated nutrient deficiency dataset using the following architecture:

- **Backbone:** MobileNetV3-Small / EfficientNet-B0 (pre-trained on ImageNet)
- **Feature Fusion Layer:** Concatenation of CNN-extracted deep features with engineered colour/texture features
- **Classification Head:** Dense layers → Softmax over deficiency categories (Healthy / N-deficiency / P-deficiency / K-deficiency / Mg-deficiency / Ca-deficiency)
- **Regression Head:** Parallel dense layers → Sigmoid activation for nutrient stress scores (0–100% per nutrient)
- **Training:** Cross-entropy loss (classification) + MSE (regression), Adam optimizer, learning rate 1e-4 with cosine annealing, batch size 32, 50 epochs with early stopping

3. Hyperparameter Optimization:

- Bayesian optimization using Optuna framework with 5-fold stratified cross-validation
- Search space: learning rate, dropout rate, number of dense layers, feature fusion strategy, augmentation intensity
- Objective: maximize weighted F1 score across deficiency classes

4. Model Interpretability:

- **Grad-CAM:** Generate heatmaps showing which leaf regions most influence each deficiency prediction
- **SHAP:** Feature importance analysis across engineered and CNN-extracted features
- **Confusion Matrix:** Detailed per-class analysis with attention to disease/deficiency symptom overlap

5. Mobile Optimization:

- TensorFlow Lite conversion with INT8 post-training quantization
- Model pruning to achieve <10 MB model size
- Benchmark inference latency on representative Android devices (target: <2 seconds per image)

Tools and Platforms

Tool/ Platform	Purpose	Justification
TensorFlow / Keras	Deep learning model development and training	Industry-standard with comprehensive TFLite support
Optuna	Bayesian hyperparameter optimization	Efficient, Python-native, supports pruning of unpromising trials
OpenCV	Image preprocessing and feature extraction	Robust computer vision library with comprehensive colour/texture tools
scikit-learn	Feature engineering and evaluation metrics	Standard ML library for feature pipelines and metrics
TensorFlow Lite	Mobile model deployment	Official mobile optimization framework, INT8 quantization support
Flutter	Mobile application development	Cross-platform (Android/iOS), single codebase
Flask/ FastAPI	API backend development	Lightweight, Python-native, ideal for ML model serving
Firebase	Real-time database & authentication	Scalable, free tier available, mobile-optimized
Git/GitHub	Version control	Industry standard for collaborative development
LabelStudio	Image annotation	Open-source, supports multi-label and bounding box annotation

Data Collection and Ethical Considerations

Data Sources:

- **Primary Source:** 40–50 FCRDI partner onion farms covering seasons 2024–2026
- **Variables:** Variety, planting date, daily weather, irrigation and fertilizer logs, disease/pest indices, image-based nutrient stress scores, yield, bulb grading, and storage details
- **Image Collection:** Onion leaf images capturing healthy leaves and N, P, K, Mg, and Ca deficiency symptoms at varying severity levels
- **Collection Method:** Farmer and extension officer entries via digital forms and the mobile app, plus key updates (e.g., fertilizer, irrigation, harvest) via SMS
- **Timeline:** Retrospective records from 2024–2025 together with prospective data from 2026, spanning at least three cropping cycles

Labelling and Verification:

- FCRDI agronomists verify deficiency type and severity through visual inspection and, where available, correlation with soil/tissue test results
- Expert panel review by Dr. W.M.K Fernando for accuracy assurance

Ethical Considerations:

- This project requires ethical clearance (submitted to Research Ethics Committee)
- Informed consent forms in Sinhala for all farmer participants
- All farmer data anonymized; yield, image, and financial details stored securely
- Data used only for research model development and evaluation
- Compliance with Sri Lankan data protection regulations

Validation Strategy and Performance Metrics

Metric	Target	Method
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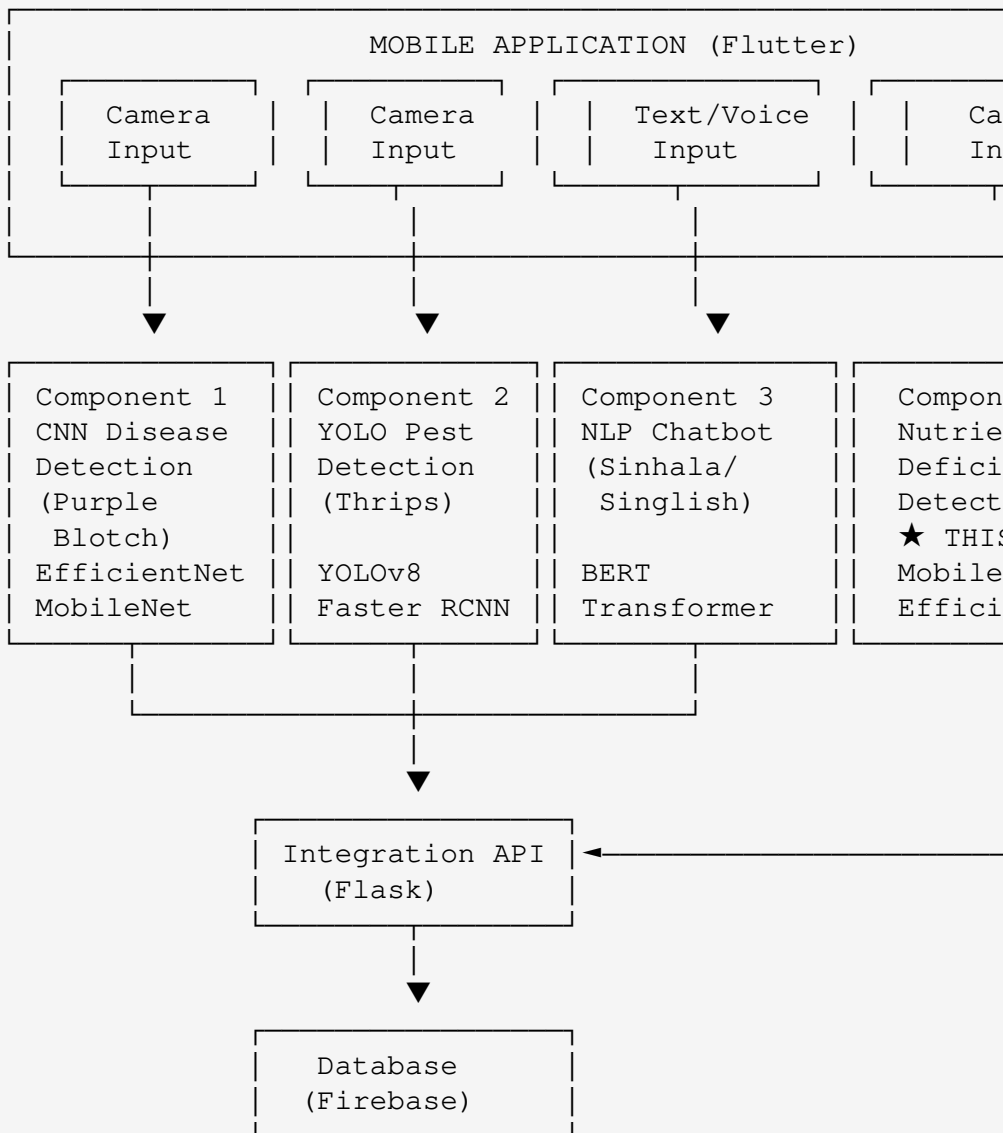
Classification Accuracy	≥ 88%	Holdout test set from unseen farms
F1 Score (macro-averaged)	≥ 85%	Precision-recall analysis per deficiency class
Stress Score MAE	≤ 10%	Mean Absolute Error on regression outputs
Grad-CAM Region Accuracy	≥ 80%	Expert agreement on highlighted regions
Model Size	< 10 MB	TFLite model after quantization/pruning
Inference Latency	< 2 seconds	Benchmarked on mid-range Android device
Fertilizer Recommendation Accuracy	≥ 80%	Expert agreement with generated recommendations
Field Validation Acceptance	≥ 75%	Farmer satisfaction survey (20+ farmers)

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of the Proposed Solution

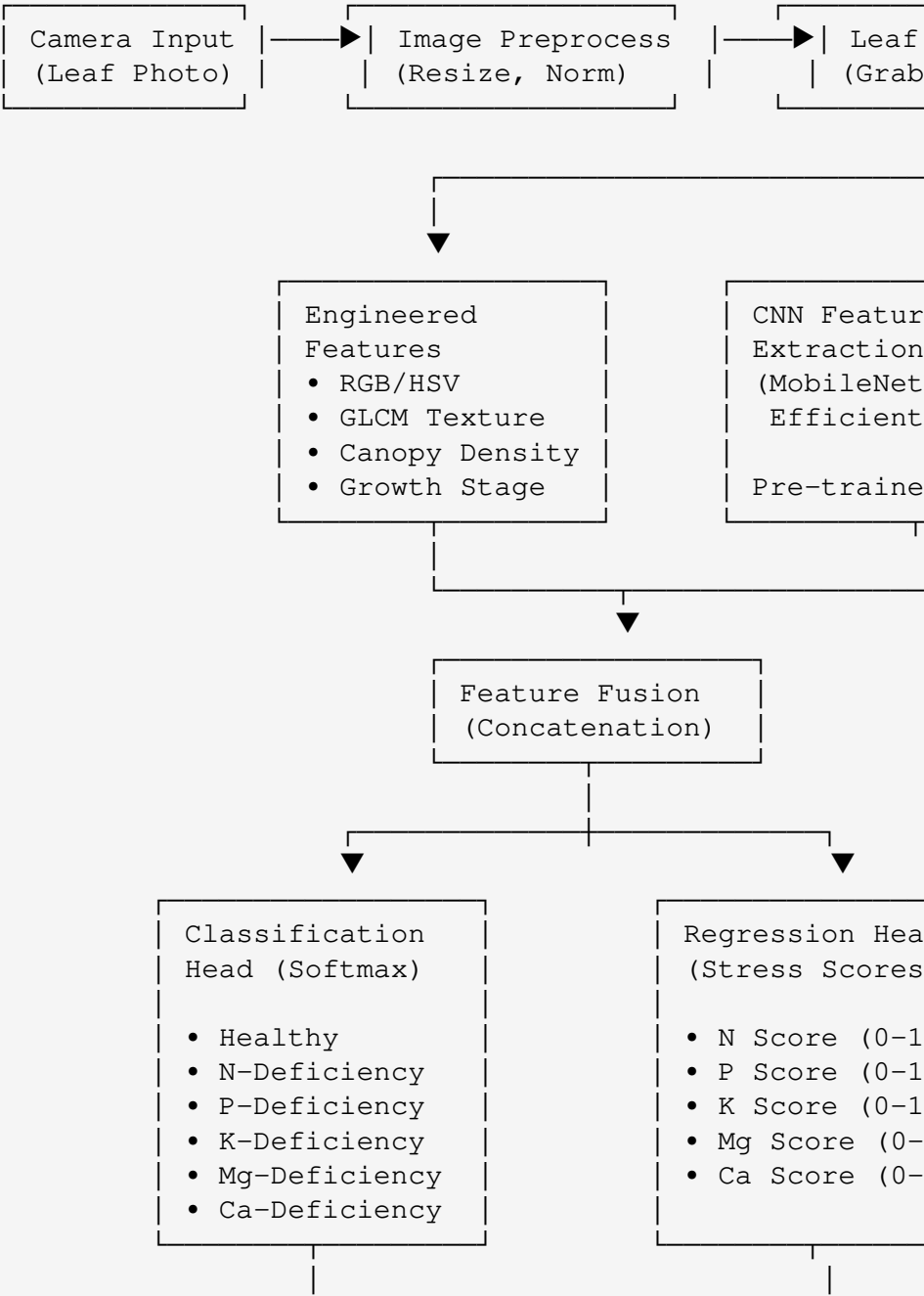
The Automated Diagnostic Framework for Onion Crop Health Monitoring is a mobile-first AI platform comprising four integrated components. This section describes the overall system and highlights the Image-Based Nutrient Deficiency Detection module (Component 4) within the larger architecture.

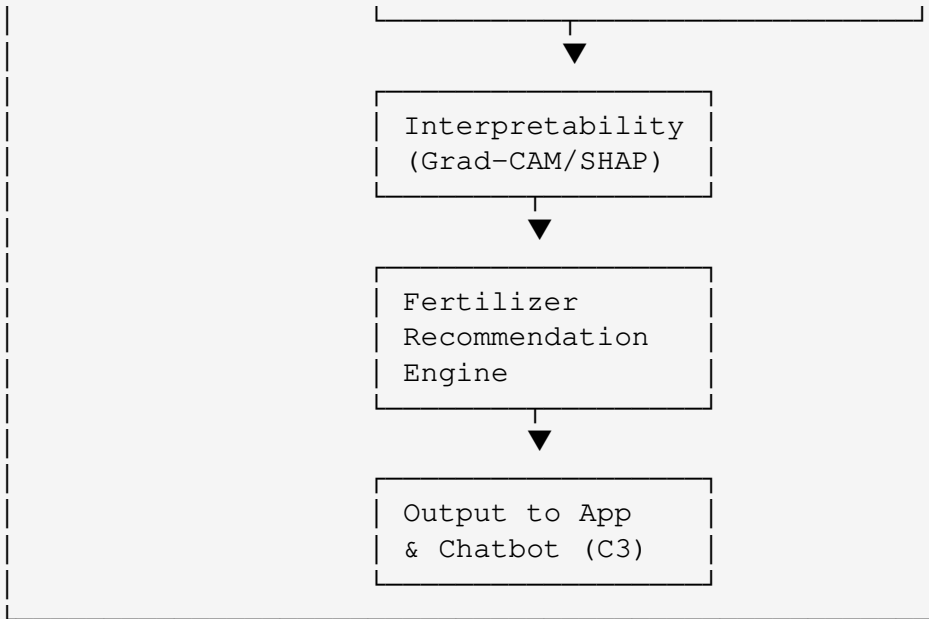
System Diagram of Components



Detailed Nutrient Deficiency Detection Architecture (Component 4)

NUTRIENT DEFICIENCY DETECTION PIPELINE





Component Interactions

The Nutrient Deficiency Detection module (Component 4) interacts with other system components through well-defined REST APIs:

- **Component 4 → Component 3:** Nutrient deficiency detection outputs (N/P/K/Mg/Ca status, severity levels, quantitative stress scores) are sent to the chatbot API. The chatbot transforms these into farmer-friendly fertilizer application guidance in Sinhala, English, or Singlish.
- **Component 1 ↔ Component 4:** Disease detection results (purple blotch severity) are shared to enable differential diagnosis —distinguishing nutrient-related discoloration from disease-related lesions on the same leaf image.
- **Component 2 ↔ Component 4:** Pest detection results (Thrips damage indicators) are shared to account for pest-induced leaf damage when assessing nutrient status, reducing false positives.

Data Flow

1. Farmer captures an onion leaf image via the mobile app camera
2. Image is preprocessed: resized, normalized, and leaf region segmented
3. Dual-path feature extraction: CNN deep features + engineered colour/texture features
4. Features are fused and passed through classification and regression heads
5. Deficiency type and quantitative stress scores are generated
6. Grad-CAM/SHAP visualisations are produced for interpretability
7. Fertilizer recommendation engine maps stress scores to actionable guidance
8. Results are displayed in the app and sent to the chatbot (Component 3) for multilingual explanation

Scalability and Security Considerations

- **Scalability:** The lightweight CNN model is designed for on-device inference, eliminating cloud dependency. The feature engineering pipeline and recommendation engine are updatable without retraining the entire model. New nutrient types or crop varieties can be added through transfer learning on additional datasets.
 - **Security:** All API communications use HTTPS encryption. Farmer data is anonymized at the point of collection. Firebase authentication ensures secure access control. Image data is stored securely and used only for model improvement with farmer consent. The system complies with Sri Lankan data protection requirements.
-

5. USER REQUIREMENTS

Stakeholder Identification

Stakeholder	Role	Primary Need
Onion Farmers	End Users	Receive accurate nutrient deficiency diagnosis and fertilizer recommendations from leaf photos
FCRDI Agronomists	Domain Experts	Validate model accuracy, ensure recommendations align with best practices
FCRDI Field Officers	Advisory Support	Monitor nutrient management outcomes, guide farmer fertilizer decisions
System Administrators	Technical Support	Manage model updates, data pipelines, and system performance

Functional Requirements

ID	Requirement	Priority
FR-01	The system shall accept onion leaf images captured via the mobile app camera	High
FR-02	The system shall preprocess images by performing leaf segmentation, resizing, and normalisation	High
FR-03	The system shall classify the input image into one of six categories: Healthy, N-deficiency, P-deficiency, K-deficiency, Mg-deficiency, Ca-deficiency	High
FR-04	The system shall generate quantitative nutrient stress scores (0–100%) for each nutrient (N, P, K, Mg, Ca)	High
FR-05	The system shall provide Grad-CAM heatmap visualisations highlighting leaf regions driving the prediction	High

FR-06	The system shall generate fertilizer recommendations with application dosage and timing guidance	High
FR-07	The system shall send deficiency results to the chatbot (Component 3) for multilingual explanation	High
FR-08	The system shall perform inference offline on the mobile device without requiring internet connectivity	High
FR-09	The system shall display a confidence score for each prediction	Medium
FR-10	The system shall log prediction history for tracking nutrient management over time	Medium
FR-11	The system shall accept optional contextual input (variety, growth stage, fertilizer history) for enhanced accuracy	Medium
FR-12	The system shall provide SHAP-based feature importance reports for expert review	Low

Table 2: Functional Requirements

Non-Functional Requirements

ID	Requirement	Category	Target
NFR-01	Inference latency shall be under 2 seconds per image on a mid-range Android device	Performance	< 2s
NFR-02	Classification accuracy shall exceed 88% on holdout test data	Accuracy	≥ 88%
NFR-03	The TFLite model size shall be under 10 MB	Efficiency	< 10 MB

NFR-04	All API communications shall use HTTPS encryption	Security	TLS 1.2+
NFR-05	Farmer image data shall be anonymized and not stored with PII	Privacy	Compliance
NFR-06	The system shall be usable by farmers with limited digital literacy	Usability	Intuitive UI
NFR-07	The model shall be updatable without full application redeployment	Maintainability	Hot model swap
NFR-08	Core deficiency detection shall work offline using the on-device model	Availability	Offline mode

Table 3: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

Primary Market: Sri Lanka's 150,000+ onion farming households concentrated in the Dry Zone (Maha Illuppallama, Matale, Dambulla, Jaffna).

Customer Persona:

- **Name:** Kumara (representative farmer)
- **Age:** 30–55 years
- **Education:** O/L or below
- **Language:** Sinhala, Singlish
- **Technology:** Owns a basic Android smartphone, limited data plan

- **Pain Point:** Cannot afford regular soil testing; relies on visual guesswork for fertilizer decisions, leading to over-fertilization and wasted input costs

Value Proposition

A single mobile application that allows farmers to photograph their onion leaves and receive instant, lab-free nutrient deficiency diagnosis with precise fertilizer recommendations—replacing costly soil testing and guesswork with AI-powered, smartphone-accessible analysis that saves money and improves yields.

Revenue Model

The system follows a **Freemium** approach:

Tier	Features	Price
Free	Basic deficiency detection (N/P/K), general fertilizer advice	Free
Premium	Full nutrient profile (N/P/K/Mg/Ca), quantitative stress scores, personalized fertilizer calendars, historical tracking	LKR 300/month per farmer
Enterprise	Bulk licensing for agricultural cooperatives, white-label options, analytics dashboard, API access	Custom pricing

Cost Estimation

Item	Estimated Annual Cost
Cloud infrastructure (GCP) – model updates and analytics	LKR 72,000
App Store + Play Store hosting	LKR 38,454

Data storage (Firebase / GCS)	LKR 12,000
Model training compute (periodic retraining)	LKR 48,000
Marketing and farmer outreach	LKR 50,000
Total	LKR 220,454

Pricing Strategy

Initial free-tier rollout to achieve adoption among 1,000+ farmers within the first 6 months. Premium features unlocked after demonstrating value through the free tier. Partnerships with fertilizer companies for co-subsidized premium access, where companies benefit from precise demand data for their products.

Competitive Advantage

1. **First onion-specific nutrient deficiency detection system** validated for Sri Lankan varieties
2. **Quantitative stress scores** replacing categorical labels
3. **Lab-free analysis** accessible via basic smartphone
4. **Smallholder-calibrated recommendations** considering budget and local fertilizer availability
5. **Offline capability** for rural areas with limited connectivity
6. **Validated by FCRDI** agricultural scientists

Intellectual Property Considerations

- The fine-tuned CNN model and curated onion nutrient deficiency dataset constitute protectable intellectual property
 - The feature fusion architecture combining engineered and deep features may be patentable
 - Open-source components (TensorFlow, Flutter) are used under their respective licenses (Apache 2.0, BSD)
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7. BUDGET AND JUSTIFICATION

Item	Description	Cost (LKR)	Occurrence	Justification
Cloud Server Hosting	GCP Compute Engine for model training and periodic retraining	6,000	Monthly	GPU-accelerated instance required for CNN training with acceptable training time
Firebase (Realtime DB + Auth)	Database hosting and user authentication	0 (Free tier)	Monthly	Free tier sufficient for initial deployment (1GB storage, 10K auth/month)
Play Store Developer Account	Publishing the mobile app on Google Play	7,754	One-Time	Required for Android app distribution
App Store Developer Account	Publishing on Apple App Store	30,700	Annual	Required for iOS app distribution
Cloud Storage (GCS)	Storing image datasets, model weights, feature logs	0.023 USD/GB	Monthly	Cost-effective cloud storage for ML artifacts and training data

Domain Name	Custom domain for API endpoints	3,000	Annual	Professional branding and consistent API access
Field Data Collection	Travel and logistics for FCRDI farm visits	15,000	One-Time	Quarterly visits to partner farms for image collection and validation
Development Hardware	No additional cost (university lab resources)	0	—	Existing lab GPUs (NVIDIA) for model training

Table 4: Budget Breakdown and Justification

Total Estimated Cost (Year 1): ~LKR 180,000

8. WORK BREAKDOWN STRUCTURE (WBS)

Task Breakdown – Component 4 (Nutrient Deficiency Detection – IT22142528)

WBS ID	Task	Duration	Deliverable
4.1	Literature review and research methodology design	Weeks 1–3	Literature review report

4.2	Compile multi-season data from 40–50 partner farms (weather, inputs, stress indices)	Weeks 3–10	Curated multi-season dataset
4.3	Engineer predictive features (GDD, colour indices, texture measures, stress markers)	Weeks 8–14	Feature engineering pipeline
4.4	Analyse key relationships between management practices and nutrient status	Weeks 12–16	Exploratory data analysis report
4.5	Train and validate base CNN classification/regression models	Weeks 14–22	Baseline trained models
4.6	Optimise stacking ensemble using Bayesian tuning (Optuna)	Weeks 20–26	Optimised model with hyperparameters
4.7	Evaluate performance on holdout set and generate SHAP/Grad-CAM explainability reports	Weeks 24–30	Evaluation report with visualisations
4.8	TFLite conversion, quantization, pruning for mobile deployment	Weeks 28–32	Mobile-optimised TFLite model (<10 MB)
4.9	Create farmer tools for nutrient status display and fertilizer recommendations	Weeks 30–36	UI components in Flutter app
4.10	Integration with Components 1, 2, and 3 APIs	Weeks 34–38	Integrated cross-component pipeline
4.11	Validate and refine the system with stakeholders in the field	Weeks 36–42	Field validation report

4.12	Iterative refinement and optimization	Weeks 40–46	Final optimised system
4.13	Documentation and thesis writing	Weeks 42–48	Final dissertation

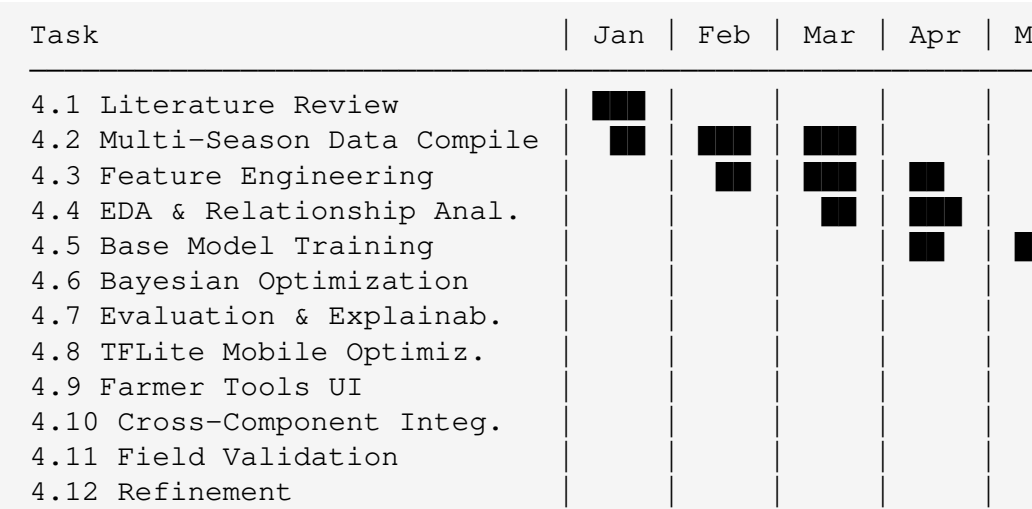
Table 5: Work Breakdown Structure

Timeline Alignment with Group

All four members follow a synchronized timeline with shared milestones at:

- **Week 12:** Data collection complete (all components)
- **Week 24:** Core models trained and individually validated
- **Week 34:** Cross-component integration complete
- **Week 42:** User testing complete
- **Week 48:** Final submission

9. GANTT CHART



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11. APPENDICES

Appendix A: Farmer Data Collection Form

Digital Data Collection Form (Sinhala/English)

1. Farm ID (auto-generated, anonymized)
2. Onion variety planted (e.g., Dambulla Red, MIII, Poona Red)
3. Planting date
4. Growth stage at image capture (Seedling / Vegetative / Bulbing / Maturity)
5. Most recent fertilizer application:
 - Type (Urea / TSP / MOP / Mixed / Other)
 - Quantity (kg per acre)
 - Date applied
6. Irrigation method and frequency
7. Any visible disease or pest symptoms? (Yes/No; if yes, describe)
8. Soil type (if known)
9. Weather conditions at capture (Sunny / Cloudy / Rainy)
10. Image capture:
 - ☐ Top of leaf (adaxial)
 - ☐ Bottom of leaf (abaxial)
 - ☐ Whole plant view

Appendix B: Sample Nutrient Deficiency Image Dataset Categories

Category	Visual Symptoms	Target Image Count
Healthy	Dark green, uniform leaves, strong upright growth	500+
Nitrogen (N) Deficiency	Pale green to yellow leaves, stunted growth, older leaves affected first	400+
Phosphorus (P) Deficiency	Dark purplish discoloration, delayed maturity, poor root development	300+
Potassium (K) Deficiency	Yellowing/browning of leaf tips, weak bulbs, marginal leaf scorch	400+

Magnesium (Mg) Deficiency	Interveinal chlorosis on older leaves, yellow patches between veins	200+
Calcium (Ca) Deficiency	Tip burn on young leaves, distorted new growth, necrotic leaf margins	200+

Total Target: 2,000+ annotated leaf images

Appendix C: Preliminary UI Mockups

(To be included – nutrient analysis interface wireframes showing camera capture, deficiency result display with stress score gauges, Grad-CAM heatmap overlay, and fertilizer recommendation panel)

Appendix D: Risk Analysis Matrix

Risk	Probability	Impact	Mitigation
Insufficient onion-specific training data	Medium	High	Augment with transfer learning from related crop datasets; apply aggressive data augmentation; collaborate with FCRDI for additional image collection
Symptom overlap between deficiency and disease	High	High	Integrate with Component 1 (disease detection) outputs for differential diagnosis; train on mixed-symptom images
Model accuracy below target on field images	Medium	High	Ensemble multiple architectures; Bayesian optimization; iterative retraining with field validation feedback

TFLite model too large for basic smartphones	Low	Medium	Progressive pruning and quantization; split model architecture if necessary
Inconsistent image quality from farmer devices	High	Medium	Implement image quality checks in the app; provide capture guidance prompts; train on diverse camera/lighting conditions
Limited farmer adoption and trust	Medium	Medium	Provide Grad-CAM visual explanations; conduct co-design sessions with farmers; validate against soil test results

Appendix E: Ethical Clearance Documents

(To be attached)

- Informed consent form (Sinhala version)
- Data management plan
- Confidentiality agreement template
- Ethics committee application form (submitted January 2026)